

A Disproof of Conjecture 00000000154

Prepared for the account `idealistic hacker`

Abstract

The conjecture claims that infinitely many even-indexed derangement numbers are prime. We prove the stronger result that no even-indexed derangement number is prime. A separate Lean 4 project formalizes the statement on actual fixed-point-free permutations and proves that the relevant index set is empty.

Frozen statement and scope

The organizer source is identified below and was frozen before this analysis:

Source: `conjectures/00000000154.md`
Commit: `95acb520ec5607c826b8a997b1ef2fc82d6f7c57`

It states:

“Prime values of derangement numbers: there exist infinitely many even n such that D_n is prime (D_n can be odd only when n is even).”

Git blob: `5d2fb2e814a4c521d2ace36034863cc8ec6fe836`
SHA-256: `694c389d34ad78b41caae8ad4e1f414fa0541e6b19b81aeab432466039dc87c1`

Here D_n has its standard meaning: the number of fixed-point-free permutations of an n -element set.

The parenthetical observation about when D_n can be odd is not a hypothesis or a conclusion of the present proof. We prove only the negation of the source’s asserted infinite set of *even* indices with prime D_n .

Derangement recurrence

The standard initial values and recurrence are

$$D_0 = 1, \quad D_1 = 0, \quad D_{r+2} = (r+1)(D_{r+1} + D_r) \quad (r \geq 0).$$

In particular, $D_2 = (1)(D_1 + D_0) = 1$.

Lemma 1. *For every $r \geq 0$, $D_{r+2} > 0$.*

Proof. Induct on r . The base case is $D_2 = 1 > 0$. If $D_{r+2} > 0$, then

$$D_{r+3} = (r+2)(D_{r+2} + D_{r+1}) > 0,$$

since $r+2 > 0$, $D_{r+2} > 0$, and derangement numbers are nonnegative. □

Theorem 1. *For every even natural number n , D_n is not prime.*

Proof. Write $n = 2k$. There are three branches.

If $k = 0$, then $n = 0$ and $D_n = D_0 = 1$, not prime. If $k = 1$, then $n = 2$ and $D_n = D_2 = 1$, also not prime. Finally suppose $k \geq 2$, so $n \geq 4$. The recurrence gives

$$D_n = (n - 1)(D_{n-1} + D_{n-2}).$$

The first factor obeys $n - 1 \geq 3$. Also $n - 2 \geq 2$ and $n - 1 \geq 2$, so the lemma gives $D_{n-2} > 0$ and $D_{n-1} > 0$. Thus $D_{n-1} + D_{n-2} \geq 2$. This expresses D_n as a product of two natural numbers both greater than one; it cannot be prime. $\square$

Thus

$$\{n \in \mathbb{N} : n \text{ is even and } D_n \text{ is prime}\} = \emptyset,$$

which is certainly not infinite. This disproves the conjecture.

Lean statement bridge

The Lean source does not formalize an unrelated recurrence alone. Mathlib’s `derangements (Fin n)` is the subtype of fixed-point-free permutations of `Fin n`; `Fintype.card` counts that actual subtype. The Mathlib theorem `card_derangements_fin_eq_numDerangements` connects this cardinality to the recurrence above.

The final Lean statements are:

$$\forall n \in \mathbb{N}, \quad \text{Even}(n) \rightarrow \neg \text{Prime}(\# \text{derangements (Fin n)}),$$

and equality of the set

$$\{n : \text{Even}(n) \wedge \text{Prime}(\# \text{derangements (Fin n)})\}$$

with the empty set, followed by its non-infinitude. Hence the formalization contains an explicit bridge from the all-even non-primality theorem to the exact set-existence wording of the source conjecture.

`Check.lean` prints the transitive axiom dependencies of every advertised theorem. The elementary recurrence lemmas print `[propext, Quot.sound]`; the statements about actual derangement cardinalities, the all-even bridge, emptiness, and non-infinitude print `[propext, Classical.choice, Quot.sound]`. These standard Lean/Mathlib principles are disclosed rather than called zero axioms; no custom axiom, `sorry`, `admit`, `native_decide`, or unsafe evaluation is used.

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